A. Let $\boldsymbol{\alpha} : I \to \mathbb{R}^3$ be given by $\boldsymbol{\alpha}(u) = \begin{cases} (u, u^2, 0) \text{ if } x > 0 \\ (u, 0, u^2) \text{ if } x \leq 0 \end{cases}$. First, we show that $\boldsymbol{\alpha}$ is always regular. When $x > 0$, $\dot{\boldsymbol{\alpha}} = (1, 2u, 0)$. When $x \leq 0$, $\dot{\boldsymbol{\alpha}} = (1, 0, 2u)$. Therefore $\boldsymbol{\alpha}$ is regular on its entire domain. Note that $\boldsymbol{T} = \frac{\dot{\boldsymbol{\alpha}}}{|\dot{\boldsymbol{\alpha}}|}$. When $x > 0$, we compute

$$\boldsymbol{T} = \frac{1}{(4u^2+1)^{1/2}} \begin{bmatrix} 1 \\ 2u \\ 0 \end{bmatrix}.$$

Note that $\boldsymbol{N} = \frac{\boldsymbol{T'}}{|\boldsymbol{T'}|}$ also compute that

$$\boldsymbol{N} = \frac{1}{(4u^2+1)^{1/2}} \begin{bmatrix} -2u \\ 1 \\ 0 \end{bmatrix}.$$

We compute $\boldsymbol{B} = \boldsymbol{T} \times \boldsymbol{N}$ as

$$\boldsymbol{B} = (0+0, 0+0, \frac{1}{4u^2+1} + \frac{4u^2}{4u^2+1}) = (0, 0, 1).$$

By observation, $\frac{d\boldsymbol{B}}{ds} = 0$. $\boldsymbol{N}$ is non-zero when $x > 0$, and therefore $\tau = 0$. When $x \leq 0$, we compute

$$\boldsymbol{T} = \frac{1}{(4u^2+1)^{1/2}} \begin{bmatrix} 1 \\ 0 \\ 2u \end{bmatrix}.$$

We also compute that

$$\boldsymbol{N} = \frac{1}{(4u^2+1)^{1/2}} \begin{bmatrix} -2u \\ 0 \\ 1 \end{bmatrix}.$$

Next, we compute $\boldsymbol{B}$.

$$\boldsymbol{B} = (0-0, \frac{-4u^2-1}{4u^2+1}, 0-0) = (0, -1, 0).$$

Again, by observation, $\frac{d\boldsymbol{B}}{ds} = 0$, and $\boldsymbol{N}$ is non-zero when $x \leq 0$, so $\tau = 0$. We can see that when $x > 0$, $\boldsymbol{\alpha}$ lies completely in the xy-plane as it has no z component, and when $x \leq 0$, $\boldsymbol{\alpha}$ lies completely in the xz-plane, as it has no y component.

B. By assumption, for every point $\boldsymbol{\alpha}(s)$, we have

$$\boldsymbol{p} = \boldsymbol{\alpha}(s) + v(s)\boldsymbol{N}(s).$$

Differentiating, we obtain

$$0 = \dot{\boldsymbol{\alpha}} + \dot{v}\boldsymbol{N} + \dot{\boldsymbol{N}}v.$$

Because $\boldsymbol{\alpha}$ is arc-length parameterized, $\dot{\boldsymbol{\alpha}} = \boldsymbol{T}$. The Serret-Frenet equations also tell us that $\frac{d\boldsymbol{N}}{ds} = -\kappa\boldsymbol{T} - \tau\boldsymbol{B}$. Substituting, we get that

$$0 = \boldsymbol{T} + \dot{v}\boldsymbol{N} + v(-\kappa\boldsymbol{T} - \tau\boldsymbol{B}).$$

Combining like terms, we get

$$0 = (1 - v\kappa)\boldsymbol{T} + \dot{v}\boldsymbol{N} - v\tau\boldsymbol{B}.$$

After much struggle, we note that $\boldsymbol{T}, \boldsymbol{N}\text{and}\boldsymbol{B}$ are orthogonal, and thus the only the only way a linear combination of them sum to 0 is if each of their coefficients are 0. Therefore, $\dot{v} = 0$, and v is constant. Also, $1 - v\kappa = 0 \implies \kappa = \frac{1}{v}$. By assumption, $\kappa \neq 0$ everywhere, and therefore $\tau = 0$. Indeed, we have shown the curvature to be constant, the torsion to be 0. Next, consider $|p - \boldsymbol{\alpha}| = |vN|$. We know that $|N| = 1$, so $|p - \boldsymbol{\alpha}| = |v|$, which is constant. Now, we have shown that α is a constant distance away from p, making it a sphere, which has torsion 0 and constant curvature. Thus, we are done.